

Unit I

LOGIC

Course Objective – Cob1: To equip students with the ability to deal with abstraction by teaching propositional logic.

Course Outcome – CO1: Apply propositional logic in practical problems.

Bloom's Taxonomy Level: K1, K2, K3, K4

Part A

S.No.	Question	BT L*																																				
1.	Define tautology and contradiction statements. Tautology: A Statement is said to be Tautology, if it is Universally True for all possible combinations of truth values. Contradiction: A Statement is said to be Contradiction, if it is Universally False for all possible combinations of truth values.	K1																																				
2.	Draw truth table for conditional, bi-conditional statements <table border="1"><thead><tr><th colspan="3">Conditional statement</th><th colspan="3">Bi-conditional statement</th></tr><tr><th><i>P</i></th><th><i>Q</i></th><th><i>P</i> → <i>Q</i></th><th><i>P</i></th><th><i>Q</i></th><th><i>P</i> ↔ <i>Q</i></th></tr></thead><tbody><tr><td>T</td><td>T</td><td>T</td><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>T</td><td>T</td><td>F</td><td>T</td><td>F</td></tr><tr><td>F</td><td>F</td><td>T</td><td>F</td><td>F</td><td>T</td></tr></tbody></table>	Conditional statement			Bi-conditional statement			<i>P</i>	<i>Q</i>	<i>P</i> → <i>Q</i>	<i>P</i>	<i>Q</i>	<i>P</i> ↔ <i>Q</i>	T	T	T	T	T	T	T	F	F	T	F	F	F	T	T	F	T	F	F	F	T	F	F	T	K1
Conditional statement			Bi-conditional statement																																			
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T	F	F	T	F	F																																	
F	T	T	F	T	F																																	
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3.	State DeMorgan’s law, Associative law, Distributive law and Commutative law with respect to ∧ and ∨. <table border="1"><thead><tr><th colspan="2">DeMorgan’s law</th><th colspan="2">Associative law</th></tr></thead><tbody><tr><td>(i) $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$</td><td>(ii) $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$</td><td>(i) $P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$</td><td>(ii) $P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$</td></tr><tr><th colspan="2">Distributive law</th><th colspan="2">Commutative law</th></tr><tr><td>(i) $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$</td><td>(ii) $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$</td><td>(i) $P \wedge Q \Leftrightarrow Q \wedge P$</td><td>(ii) $P \vee Q \Leftrightarrow Q \vee P$</td></tr></tbody></table>	DeMorgan’s law		Associative law		(i) $\neg(P \wedge Q) \Leftrightarrow \neg P \vee \neg Q$	(ii) $\neg(P \vee Q) \Leftrightarrow \neg P \wedge \neg Q$	(i) $P \wedge (Q \wedge R) \Leftrightarrow (P \wedge Q) \wedge R$	(ii) $P \vee (Q \vee R) \Leftrightarrow (P \vee Q) \vee R$	Distributive law		Commutative law		(i) $P \wedge (Q \vee R) \Leftrightarrow (P \wedge Q) \vee (P \wedge R)$	(ii) $P \vee (Q \wedge R) \Leftrightarrow (P \vee Q) \wedge (P \vee R)$	(i) $P \wedge Q \Leftrightarrow Q \wedge P$	(ii) $P \vee Q \Leftrightarrow Q \vee P$	K1																				
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4.	State Identity law, Negation law, Double Negation law and Idempotent law with respect to ∧ and ∨. <table border="1"><thead><tr><th colspan="2">Identity law</th><th colspan="2">Negation law</th></tr></thead><tbody><tr><td>(i) $P \wedge T \Leftrightarrow P$</td><td>(ii) $P \vee F \Leftrightarrow P$</td><td>(i) $P \wedge \neg P \Leftrightarrow F$</td><td>(ii) $P \vee \neg P \Leftrightarrow T$</td></tr><tr><th colspan="2">Double Negation law</th><th colspan="2">Idempotent law</th></tr><tr><td>(i) $\neg(\neg P) \Leftrightarrow P$</td><td></td><td>(i) $P \wedge P \Leftrightarrow P$</td><td>(ii) $P \vee P \Leftrightarrow P$</td></tr></tbody></table>	Identity law		Negation law		(i) $P \wedge T \Leftrightarrow P$	(ii) $P \vee F \Leftrightarrow P$	(i) $P \wedge \neg P \Leftrightarrow F$	(ii) $P \vee \neg P \Leftrightarrow T$	Double Negation law		Idempotent law		(i) $\neg(\neg P) \Leftrightarrow P$		(i) $P \wedge P \Leftrightarrow P$	(ii) $P \vee P \Leftrightarrow P$	K1																				
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5.	What is the contrapositive, the converse and the inverse of the conditional statement. “If you work hard then you will be rewarded.” <table><tr><td>Given that</td><td>$P \rightarrow Q$</td><td>“If you work hard then you will be rewarded.”</td></tr><tr><td>Contrapositive</td><td>$\neg Q \rightarrow \neg P$</td><td>“If you are not rewarded then you will not work hard.”</td></tr><tr><td>Converse</td><td>$Q \rightarrow P$</td><td>“If you will be rewarded then you work hard.”</td></tr><tr><td>Inverse</td><td>$\neg P \rightarrow \neg Q$</td><td>“If you are not work hard then you will not be rewarded.”</td></tr></table>	Given that	$P \rightarrow Q$	“If you work hard then you will be rewarded.”	Contrapositive	$\neg Q \rightarrow \neg P$	“If you are not rewarded then you will not work hard.”	Converse	$Q \rightarrow P$	“If you will be rewarded then you work hard.”	Inverse	$\neg P \rightarrow \neg Q$	“If you are not work hard then you will not be rewarded.”	K2													
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6.	Construct a truth table for the compound proposition $(P \rightarrow Q) \rightarrow (Q \rightarrow P)$ <table><tr><td>P</td><td>Q</td><td>$P \rightarrow Q$</td><td>$Q \rightarrow P$</td><td>$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$</td></tr><tr><td>T</td><td>T</td><td>T</td><td>T</td><td>T</td></tr><tr><td>T</td><td>F</td><td>F</td><td>T</td><td>T</td></tr><tr><td>F</td><td>T</td><td>T</td><td>F</td><td>F</td></tr><tr><td>F</td><td>F</td><td>T</td><td>T</td><td>T</td></tr></table>	P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$	T	T	T	T	T	T	F	F	T	T	F	T	T	F	F	F	F	T	T	T	K2
P	Q	$P \rightarrow Q$	$Q \rightarrow P$	$(P \rightarrow Q) \rightarrow (Q \rightarrow P)$																							
T	T	T	T	T																							
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7.	State some laws of inference theory: Disjunctive Syllogism, Modus Ponens, Modus Tollens, Hypothetical syllogism and Dilemma. <table><tr><td>Disjunctive Syllogism</td><td>$\neg P, P \vee Q \Rightarrow Q$</td></tr><tr><td>Modus Ponens</td><td>$P, P \rightarrow Q \Rightarrow Q$</td></tr><tr><td>Modus Tollens</td><td>$\neg Q, P \rightarrow Q \Rightarrow \neg P$</td></tr><tr><td>Hypothetical syllogism</td><td>$P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$</td></tr><tr><td>Dilemma</td><td>$P \vee Q, P \rightarrow R, Q \rightarrow R \Rightarrow R$</td></tr></table>	Disjunctive Syllogism	$\neg P, P \vee Q \Rightarrow Q$	Modus Ponens	$P, P \rightarrow Q \Rightarrow Q$	Modus Tollens	$\neg Q, P \rightarrow Q \Rightarrow \neg P$	Hypothetical syllogism	$P \rightarrow Q, Q \rightarrow R \Rightarrow P \rightarrow R$	Dilemma	$P \vee Q, P \rightarrow R, Q \rightarrow R \Rightarrow R$	K1															
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8.	Define universal, existential quantifiers and its rules Universal quantifiers: Suppose “all elements of the domain satisfy a certain property of the predicate $P(x)$, then we denote it by $(\forall x), P(x)$ which means “$P(x)$ is true for every x” Existential quantifiers: Suppose “there exists some elements (or) at least one element of the domain satisfy a certain property of the predicate $P(x)$, then we denote it by $(\exists x), P(x)$ which means “$P(x)$ is true for some x” Rules of Quantifiers(Specification): <table><tr><td>Quantifiers</td><td>Rules</td><td>Predicates</td></tr><tr><td>$(\forall x), P(x)$</td><td>Rule US (Universal Specification)</td><td>$P(\alpha), \alpha$ is arbitrary</td></tr><tr><td>$(\exists x), P(x)$</td><td>Rule ES (Existential Specification)</td><td>$P(a),$ for some a</td></tr></table> Rules of Quantifiers(Generalisation): <table><tr><td>Predicates</td><td>Rules</td><td>Quantifiers</td></tr><tr><td>$P(\alpha), \alpha$ is arbitrary</td><td>Rule UG (Universal Generalisation)</td><td>$(\forall x), P(x)$</td></tr><tr><td>$P(a),$ for some a</td><td>Rule EG (Existential Generalisation)</td><td>$(\exists x), P(x)$</td></tr></table>	Quantifiers	Rules	Predicates	$(\forall x), P(x)$	Rule US (Universal Specification)	$P(\alpha), \alpha$ is arbitrary	$(\exists x), P(x)$	Rule ES (Existential Specification)	$P(a),$ for some a	Predicates	Rules	Quantifiers	$P(\alpha), \alpha$ is arbitrary	Rule UG (Universal Generalisation)	$(\forall x), P(x)$	$P(a),$ for some a	Rule EG (Existential Generalisation)	$(\exists x), P(x)$	K1							
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9.	Write the symbolic form of the statement, (i). “Some cats have no tails.” Solution: Let us consider the following symbols $C(x)$: x is a cat $T(x)$: x has tail The Symbolic representation of “Some cats have no tails.” is $(\exists x), C(x) \wedge \neg T(x)$ (ii). “All men are giants” & “Some men are giants”. Solution: Let us consider the following symbols $M(x)$: x is a Man $G(x)$: x is a Giants The Symbolic representation of “All men are giants” is $(\forall x), M(x) \rightarrow G(x)$ The Symbolic representation of “Some men are giants” is $(\exists x), M(x) \wedge G(x)$ (iii). Socrates is a mortal is a valid argument from the premises All men are mortal and Socrates is a man Solution: Let us consider the following symbols $H(x)$: x is a Man $m(x)$: x is a mortal s : s is a Socrates The Symbolic representation of “All men are mortal” is $(\forall x), H(x) \rightarrow m(x)$ The Symbolic representation of Socrates is a man is $H(s)$ Now the given premises are $P_1: (\forall x), H(x) \rightarrow m(x)$ $P_2: H(s)$ <table border="1" data-bbox="301 1193 1335 1476"> <thead> <tr> <th>Steps</th><th>Arguments</th><th>Reasons</th></tr> </thead> <tbody> <tr> <td>1</td><td>$(\forall x), H(x) \rightarrow m(x)$</td><td>Rule P</td></tr> <tr> <td>2</td><td>$H(\alpha) \rightarrow m(\alpha)$</td><td>Rule US ($\alpha$ is arbitrary)</td></tr> <tr> <td>3</td><td>$H(s)$</td><td>Rule P</td></tr> <tr> <td>4</td><td>$m(s)$</td><td>Rule T (Modus Ponens)</td></tr> <tr> <td>Conclusion</td><td>Socrates is a mortal</td><td></td></tr> </tbody> </table>	Steps	Arguments	Reasons	1	$(\forall x), H(x) \rightarrow m(x)$	Rule P	2	$H(\alpha) \rightarrow m(\alpha)$	Rule US (α is arbitrary)	3	$H(s)$	Rule P	4	$m(s)$	Rule T (Modus Ponens)	Conclusion	Socrates is a mortal		K2
Steps	Arguments	Reasons																		
1	$(\forall x), H(x) \rightarrow m(x)$	Rule P																		
2	$H(\alpha) \rightarrow m(\alpha)$	Rule US (α is arbitrary)																		
3	$H(s)$	Rule P																		
4	$m(s)$	Rule T (Modus Ponens)																		
Conclusion	Socrates is a mortal																			
10.	Verify the validity of the following argument. All mathematics professors have studied number theory. Leo is a mathematics professor. Therefore Leo has studied number theory. Solution: Let us consider the following symbols $M(x)$: x is a Mathematics Professor $N(x)$: x have studied Number Theory L : L is a Leo The Symbolic representation of “All mathematics professors have studied number theory” is $(\forall x), M(x) \rightarrow N(x)$ The Symbolic representation of Leo is a mathematics professor is $M(L)$ Now the given premises are $P_1: (\forall x), M(x) \rightarrow N(x)$ $P_2: M(L)$	K2																		

Steps	Arguments	Reasons
1	$(\forall x), M(x) \rightarrow N(x)$	Rule P
2	$M(\alpha) \rightarrow N(\alpha)$	Rule US (α is arbitrary)
3	$M(L)$	Rule P
4	$N(L)$	Rule T (Modus Ponens)
Conclusion	Leo has studied Number Theory	

Part B

S.N	Question	BT L
1.	Show that the followings are tautology by using equivalence relations (i). $((P \vee Q) \wedge \neg(\neg P \wedge (\neg Q \vee \neg R))) \vee (\neg P \wedge \neg Q) \vee (\neg P \wedge \neg R)$ (ii). $[(p \vee q) \wedge (\neg p \vee r) \wedge (\neg q \vee r)] \rightarrow r$	K4
2.	Prove that the following equivalence relations without using Truth table (i) $(P \rightarrow R) \wedge (Q \rightarrow R) \Leftrightarrow (P \vee Q) \rightarrow R$ (ii) $[(P \wedge Q \wedge R) \rightarrow S] \wedge [R \rightarrow (P \vee Q \vee S)] \Leftrightarrow [R \wedge (P \leftrightarrow Q)] \rightarrow S$ (iii). $[(P \wedge Q) \rightarrow R] \wedge [Q \rightarrow (S \vee R)] \Leftrightarrow [Q \wedge (S \rightarrow P)] \rightarrow R$	K4
3.	Obtain the PDNF and PCNF using equivalence relations of followings (i). $(\neg P \rightarrow R) \wedge (Q \leftrightarrow P)$ (ii). $(P \wedge Q) \vee (\neg P \wedge R)$ (iii). $[q \vee (p \wedge r)] \wedge \neg[(p \vee r) \wedge q]$ (iv). $[p \rightarrow (q \wedge p)] \wedge [\neg p \rightarrow (\neg q \wedge \neg r)]$	K4
4.	Prove the following argument are valid (i). $R \rightarrow S$ is logically derived from the premises $P \rightarrow (Q \rightarrow S)$, $\neg R \vee P$ and Q (ii). $P \rightarrow \neg S$ from $P \rightarrow (Q \vee R)$, $Q \rightarrow \neg P$, $S \rightarrow \neg R$ and P (by indirect method) (iii). $(P \rightarrow Q) \wedge (R \rightarrow S)$, $(Q \rightarrow M) \wedge (S \rightarrow N)$, $\neg(M \wedge N)$ and $(P \rightarrow R) \Rightarrow \neg P$ (iv). $R \vee S$ follows logically from the premises $C \vee D$, $(C \vee D) \rightarrow \neg H$, $\neg H \rightarrow (A \wedge \neg B)$ and $(A \wedge \neg B) \rightarrow (R \vee S)$.	K4
5.	(i). Show that the hypotheses, “It is not sunny this afternoon and it is colder than yesterday.” “We will go swimming only if it is sunny.” “If we do not go swimming, then we will take a canoe trip.” and “If we take a canoe trip, then we will go home by sunset” lead to the conclusion “We will be home by sunset.” (ii) Show that the following premises are inconsistent: a. If Jack misses many classes through illness, then he fails high school. b. If Jack fails high school, then he is uneducated. c. If Jack reads a lot of books, then he is not uneducated. d. Jack misses many classes through illness and reads a lot of books.	K4
6.	Show that (i) $(\forall x)[P(x) \rightarrow Q(x)] \wedge (\forall x)[Q(x) \rightarrow R(x)] \Rightarrow (\forall x)[P(x) \rightarrow R(x)]$. (ii) $(\forall x)[P(x) \vee Q(x)]$, $(\forall x)[\neg P(x) \wedge Q(x)] \rightarrow R(x) \Rightarrow (\forall x)[\neg R(x) \rightarrow P(x)]$. (iii) $(\forall x)[P(x) \rightarrow (Q(x) \wedge R(x))]$, $(\forall x)[P(x) \wedge S(x)] \Rightarrow (\forall x)[R(x) \wedge S(x)]$.	K4
7.	(i). Verify the validity of the following argument. No junior or senior is enrolled in a physical education class. Mary is enrolled in a physical education class. Therefore Mary is not a senior.	K4

	(ii). Show that the premises “One student in this class knows how to write programs in JAVA” and “Everyone who knows how to write program in JAVA can get a high paying job” imply the conclusion “Someone in this class can get a high paying job.”	
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